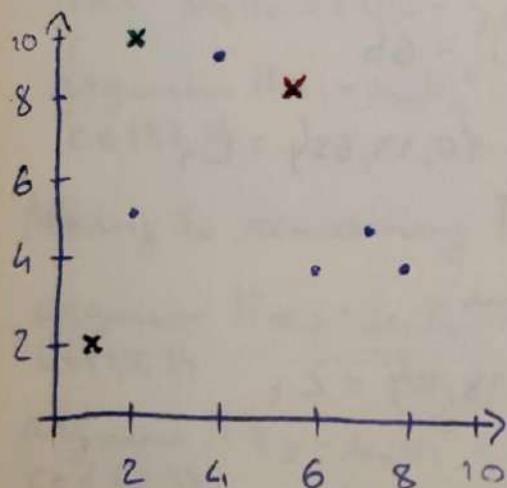


Homework 4

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Part A

a)



b)

k-means algorithm

- 1) initialize centroids
- 2) assign points to clusters
- 3) adjust centroids
- 4) re-assign points

change? $\rightarrow$ yes $\rightarrow$ go to ③
 $\rightarrow$ no $\rightarrow$ convergence

- 1) initialize centroids

$$\mu_1 = \mu_7 = [2 \ 10]^T$$

$$\mu_2 = \mu_4 = [5 \ 8]^T$$

$$\mu_3 = \mu_7 = [1 \ 2]^T$$

2) assign points to clusters

Let us start with x_1 :

$$\|x_1 - \mu_1\|_2^2 = 0$$

$$\|x_1 - \mu_2\|_2^2 = (\sqrt{(2-5)^2 + (10-8)^2})^2 = 13$$

$$\|x_1 - \mu_3\|_2^2 = (\sqrt{(2-1)^2 + (10-2)^2})^2 = 65$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_1 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{0, 13, 65\} = c_1$$

Moving to remaining observations:

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_2 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{25, 18, 10\} = c_3$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_3 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{72, 25, 53\} = c_2$$

$$x_4 \text{ is the centroid } c_2 \text{ so } \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_4 - \mu_c\|_2^2 = c_2$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_5 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{50, 13, 45\} = c_2$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_6 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{52, 17, 29\} = c_2$$

$$x_7 \text{ is the centroid } c_3 \text{ so } \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_7 - \mu_c\|_2^2 = c_3$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_8 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{5, 2, 58\} = c_2$$

3) adjust centroids

$$\mu_1 = x_1 = [2 \ 10]^T$$

$$\mu_2 = \frac{x_3 + x_4 + x_5 + x_6 + x_8}{5} = [6 \ 6]^T$$

$$\mu_3 = \frac{x_2 + x_7}{2} = [7.5 \ 3.5]^T$$

4) re-assign points

Let us start with x_1 :

$$\|x_1 - \mu_1\|_2^2 = (\sqrt{(2-2)^2 + (10-10)^2})^2 = 0$$

$$\|x_1 - \mu_2\|_2^2 = (\sqrt{(2-6)^2 + (10-6)^2})^2 = 32$$

$$\|x_1 - \mu_3\|_2^2 = (\sqrt{(2-7,5)^2 + (10-3,5)^2})^2 = 42,5$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_1 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{0; 32; 42,5\} = c_1$$

Moving to remaining observations:

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_2 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{25; 17; 2,5\} = c_3$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_3 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{72; 8; 42,5\} = c_2$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_4 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{13; 5; 32,5\} = c_2$$

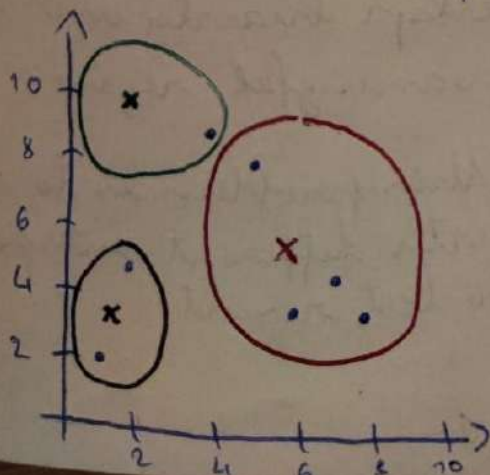
$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_5 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{50; 2; 32,5\} = c_2$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_6 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{52; 4; 20,5\} = c_2$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_7 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{65; 47; 2,5\} = c_3$$

$$\underset{c \in \{1,2,3\}}{\operatorname{argmin}} \|x_8 - \mu_c\|_2^2 = \underset{c \in \{1,2,3\}}{\operatorname{argmin}} \{5; 13; 36,5\} = c_1$$

c)



d)

The K-means algorithm is very sensitive to centroid initialization. If the initial centroids are already near the true cluster centers, then each data point is quickly assigned to the correct cluster, and the centroids do not need to move much in each iteration. Thus, the ~~algorithm~~ algorithm converges faster.

On the other hand, poor initialization can place multiple centroids in the same region or far from any data. In such cases, many points are misassigned initially, and the centroids must travel a long distance to "find" their correct clusters. As a result, the algorithm wastes time correcting earlier mistakes, converges more slowly and it might not even reach the optimal solution.

For example, if a centroid starts too far from the data points, it might not get any points assigned to it, leaving the cluster empty. Conversely, if multiple centroids start too close to one another, the resulting clusters may overlap heavily and fail to separate the data into meaningful regions.

A common way to address this problem is to run K-means multiple times with different centroid initialization and select the best result. ::

Part B

$$a) \text{var}(x) = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n-1} \quad \text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\bar{y}_1 = \frac{5+0+1}{3} = 2 \quad \bar{y}_2 = \frac{5}{3} \quad \bar{y}_3 = 0$$

$$\text{var}(y_1) = \frac{(5-2)^2 + (0-2)^2 + (1-2)^2}{2} = 7$$

$$\text{var}(y_2) = \frac{(0-5/3)^2 + (5-5/3)^2 + (0-5/3)^2}{2} = \frac{25}{3}$$

$$\text{var}(y_3) = \frac{1^2 + 0^2 + (-1)^2}{2} = 1$$

$$\text{cov}(y_1, y_2) = \frac{(5-2)(0-5/3) + (0-2)(5-5/3) + (1-2)(0-5/3)}{2} = -5$$

$$\text{cov}(y_1, y_3) = \frac{(5-2) \times 1 + (0-2) \times 0 + (1-2) \times (-1)}{2} = 2$$

$$\text{cov}(y_2, y_3) = \frac{(0-5/3) \times 1 + (5-5/3) \times 0 + (0-5/3) \times (-1)}{2} = 0$$

$$\Sigma = \begin{bmatrix} 7 & -5 & 2 \\ -5 & \frac{25}{3} & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

b)

Let's calculate the eigenvalues and eigenvectors of the covariance matrix Σ

$$\det(\Sigma - \lambda I) = 0 \Leftrightarrow \begin{vmatrix} 7-\lambda & -5 & 2 \\ -5 & \frac{25}{3}-\lambda & 0 \\ 2 & 0 & 1 \end{vmatrix} = 0 \Leftrightarrow$$

$$\Leftrightarrow (7-\lambda)\left(\frac{25}{3}-\lambda\right)(1-\lambda) + 5(-5)(1-\lambda) + 2(-2)\left(\frac{25}{3}-\lambda\right) = 0 \Leftrightarrow$$

$$\Leftrightarrow (7-\lambda)\left(\frac{25}{3}-\frac{25}{3}\lambda + \lambda^2\right) - 25 + 25\lambda - \frac{100}{3} + 4\lambda = 0 \Leftrightarrow$$

$$\Leftrightarrow \frac{175}{3} - \frac{196}{3}\lambda + 7\lambda^2 - \frac{25}{3}\lambda + \frac{28}{3}\lambda^2 - \lambda^3 - \frac{175}{3} + 29\lambda = 0 \Leftrightarrow$$

$$\Leftrightarrow -\lambda^3 + \frac{49}{3}\lambda^2 - \frac{134}{3}\lambda = 0 \Leftrightarrow$$

$$\Leftrightarrow \lambda = 0 \vee -\lambda\left(\lambda^2 - \frac{49}{3}\lambda + \frac{134}{3}\right) = 0 \Leftrightarrow$$

$$\Leftrightarrow \lambda = 0 \vee \lambda \approx 3,473297 \vee \lambda = 12,860043$$

$$\lambda_1 = 12,860043 \quad \lambda_2 = 3,473297 \quad \lambda_3 = 0$$

To determine a 2 dim. space that minimizes the projection error, we'll choose the 2 biggest eigenvalues and calculate their eigenvectors.

$$\lambda_1 = 12,860043 \quad \lambda_2 = 3,473297$$

$$\Sigma v_1 = \lambda_1 v_1 \Leftrightarrow (\Sigma - \lambda_1 I) v_1 = 0 \Leftrightarrow$$

$$\begin{bmatrix} -5,860043 & -5 & 2 \\ -5 & -4,526710 & 0 \\ 2 & 0 & -11,860043 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \\ v_{13} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow$$

$$\begin{cases} -5v_{11} - 4,526710v_{12} = 0 \\ 2v_{11} - 11,860043v_{13} = 0 \end{cases} \Rightarrow \begin{cases} v_{12} = \frac{-5}{-4,526710} v_{11} = 1,104535v_{11} \\ v_{11} = 5,930022v_{13} \end{cases}$$

assuming $v_{13} = 1 \Rightarrow v_1 = \begin{bmatrix} 5,930022 \\ -6,550035 \\ 1 \end{bmatrix}$

normalize $v_1' = \begin{bmatrix} 0,666892 \\ -0,736619 \\ 0,112460 \end{bmatrix}$

$$\Sigma v_2 = \lambda_2 v_2 \Rightarrow (\Sigma - \lambda_2 I) v_2 = 0 \Rightarrow$$

$$\begin{bmatrix} 3,526709 & -5 & 2 \\ -5 & 4,860042 & 0 \\ 2 & 0 & -2,473297 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \\ v_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$\begin{cases} -5v_{21} + 4,860042v_{22} = 0 \\ 2v_{21} - 2,473297v_{23} = 0 \end{cases} \Rightarrow \begin{cases} v_{22} = \frac{5}{4,860042} v_{21} = 1,028811v_{21} \\ v_{21} = 1,236646v_{23} \end{cases}$$

assuming $v_{23} = 1 \Rightarrow v_2 = \begin{bmatrix} 1,236646 \\ 1,272259 \\ 1 \end{bmatrix}$

normalize $v_2' = \begin{bmatrix} 0,607197 \\ 0,624683 \\ 0,497003 \end{bmatrix}$

$$U = \begin{bmatrix} 0,666892 & 0,607197 \\ -0,736619 & 0,624683 \\ 0,112460 & 0,497003 \end{bmatrix}$$

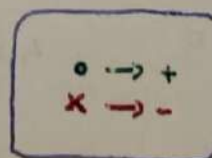
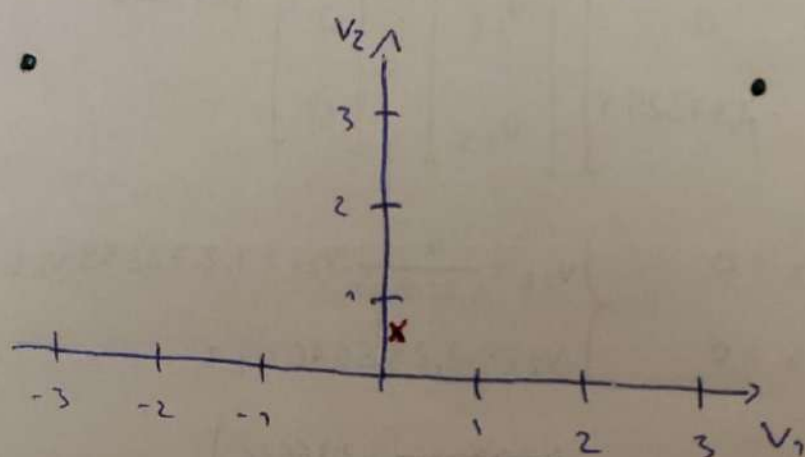
projection plane: $\text{Span}\{v_1', v_2'\}$

c)

$$\mathbf{x}_1' = \mathbf{U}^T \mathbf{x}_1 = \mathbf{U}^T \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3,446920 \\ 3,526988 \end{bmatrix}$$

$$\mathbf{x}_2' = \mathbf{U}^T \mathbf{x}_2 = \mathbf{U}^T \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} -3,683095 \\ 3,723415 \end{bmatrix}$$

$$\mathbf{x}_3' = \mathbf{U}^T \mathbf{x}_3 = \mathbf{U}^T \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0,554432 \\ 0,776794 \end{bmatrix}$$



The projection into the PCA plane shows that the "+" and "-" classes are clearly distinguishable. Although the first principal component v_1 corresponds to the direction of highest variance, the separation between the two classes occurs along the second principal component v_2 . In the plot, the points from the positive class are located higher along v_2 , while the negative class point lies lower, indicating that v_2 provides better class discrimination.